NAME : ABINEKAVARSHINI.M

CO1 - AT8

REG NO : 924260333

SUBJECT : AI

SUBJECT CODE : CSA1701

1. An intelligent agent observes its environment through sensors and acts on it through devices. In a smart home, sensors might detect motion, temperature, light and such as lights, thermostats and door locks carry out the agent's actions.

SIMPLE REFLEX AGENT :-

Acts only on the current input using condition-action rules. For example: If motion is detected after sunset, turn on the hallway light. It is fast and simple, but does not remember earlier events or adapt well to changing circumstances.

MODEL-BASED AGENT :-

Keeps an internal model of the home and uses it to interpret current sensor readings. It could remember that someone is home even if motion sensors are temporarily inactive or track that a room is warming up. This helps when information is incomplete, though the model must be kept accurate.

GOAL - BASED AGENT :

Choose actions that help achieve a stated goal, such as keeping the living room with a comfortable temperature range or ensuring the door locked at night. It can consider sequences of actions but may not know which of several goal - satisfying option is best.

UTILITY BASED AGENT

Does predict outcomes according to factors such as comfort, energy use and security, then select an action with the highest overall value. For example: it might choose a slightly warmer thermostat setting when the home is empty to save energy, while prioritizing comfort when occupants return

HYBRID APPROACH :-

It would work best. Simple reflex rules can handle urgent events such as smoke detection or an open door at night. A model - based component can track occupancy and environmental changes. Goal - based planning can support routines, while a utility based component balances comfort, energy efficiency and security. Safety rules should remain fixed for critical situations.

2. The robot can represent each location as a state available moves as actions and the exist as the goal. Because the maze is unknown it must explore, record what it observes and use that information to guide later decisions. The search algorithm assume the robot can identify its current location and remember discovered paths

UNIFORM COST SEARCH

UCS explores the path with the lowest total cost so far. It is useful when moves have different costs - for example if some paths are longer, harder to traverse or consume more battery

Advantage : Finds a least-cost path when steps are non-negative

Disadvantage : Can use substantial time and memory because it stores many frontier paths

If all moves have equal cost, UCS still finds a shortest path, but its cost-prioritized exploration may do more work than necessary.

ITERATIVE DEEPENING SEARCH

IDS repeatedly performs depth-limited depth-first searches, increasing the deep limit each time until it finds the exit

Advantage : Uses little memory and is complete when the branching factor is finite and a solution exists at a finite depth. With equal move costs, it finds a shallowest and therefore shortest solution

Disadvantage : Repeats exploration near the start at each new depth limit. It is not suited to finding a least cost path when have different costs.

AGENT DESIGN :

A simple reflex agent might choose an move based only on what is currently sensed. In an unfamiliar maze, that can lead to repeated moves or loops because it does not remember explored locations. A model based, based agent can maintain a map of known corridors, walls and visited locations. In either case, the agent must update its map when its discover a new parts of the maze